# Unified Derivation of Interband $\varepsilon_2(\hbar\omega)$

Reduction of the 3D Integral to One Dimension at the X and L Points

*Following Guerrisi, Rosei & Winsemius, Phys. Rev. B **12**, 557 (1975)*

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## 1 Starting Point: Fermi's Golden Rule

The imaginary part of the dielectric function for direct interband absorption is

$$\varepsilon_2(\hbar\omega) \;\propto\; \frac{|P|^2}{(\hbar\omega)^2}\int_{\mathrm{BZ}} \mathrm{d}^3k\; f\big(\mathcal{E}_v(\mathbf{k})\big)\,\big[1-f\big(\mathcal{E}_c(\mathbf{k})\big)\big]\;\delta\big(\mathcal{E}_c(\mathbf{k})-\mathcal{E}_v(\mathbf{k})-\hbar\omega\big), \tag{1}$$

where $f(\mathcal{E}) = [1+e^{\mathcal{E}/k_BT}]^{-1}$ is the Fermi–Dirac distribution with energies measured from $E_F = 0$, and $|P|^2$ is the (locally constant) optical dipole matrix element. We assume direct transitions ($\mathbf{k}_1 \approx \mathbf{k}_2$) so both bands are evaluated at the same $\mathbf{k}$.

### 1.1 Band Dispersions (Saddle Point at Both X and L)

Near each critical point the bands are expanded to second order with cylindrical symmetry ($k_\perp$ transverse to the symmetry axis, $k_\parallel$ along it):

$$\boxed{\mathcal{E}_v(\mathbf{k}) = -\mathcal{E}_{0v} - A_v k_\perp^2 - B_v k_\parallel^2, \qquad \mathcal{E}_c(\mathbf{k}) = \mathcal{E}_{0c} + A_c k_\perp^2 - B_c k_\parallel^2,} \tag{2}$$

with all coefficients $A_c, B_c, A_v, B_v > 0$. The conduction band is a *saddle point* at both X and L: it curves **upward** along $k_\perp$ and **downward** along $k_\parallel$. Since the sign structure is identical at both critical points, the entire derivation proceeds in general form.

Table 1: Critical-point parameters. Curvature coefficients $A_\alpha = C/m^*_{\alpha\perp}$, $B_\alpha = C/m^*_{\alpha\parallel}$ with $C \equiv \hbar^2/(2m_e) = 3.81\ \mathrm{eV\,\AA}^2$.

| Point | $N$ | $\mathcal{E}_{0c}$ | $\mathcal{E}_g$ | Topology | Symmetry axis |
|---|---|---|---|---|---|
| X ($\Delta$) | 6 | 0 (reference) | $\sim$1.94 eV | Saddle ($M_1$) | $k_\parallel$ along $\Delta$ |
| L ($\Lambda$) | 8 | $\mathcal{E}^L_{0c} \neq 0$ | $\sim$2.45 eV | Saddle ($M_1$) | $k_\parallel$ along $\Lambda$ (toward $\Gamma$) |

## 2 Momentum-to-Energy Transformation

### 2.1 Change of Variables I: Cylindrical Symmetry

Exploiting the azimuthal symmetry around the symmetry axis,

$$\mathrm{d}^3k = \mathrm{d}k_x\,\mathrm{d}k_y\,\mathrm{d}k_z \;\longrightarrow\; k_\perp\,\mathrm{d}k_\perp\,\mathrm{d}k_\parallel\,\mathrm{d}\phi \;\longrightarrow\; 2\pi\,k_\perp\,\mathrm{d}k_\perp\,\mathrm{d}k_\parallel. \tag{3}$$

### 2.2 Change of Variables II: Linearization of Momenta

Define $u \equiv k_\perp^2 \geq 0$ and $v \equiv k_\parallel^2 \geq 0$:

$$\mathrm{d}u = 2k_\perp\,\mathrm{d}k_\perp, \qquad \mathrm{d}v = 2|k_\parallel|\,\mathrm{d}k_\parallel \quad\Rightarrow\quad \mathrm{d}k_\parallel = \frac{\mathrm{d}v}{2\sqrt{v}}. \tag{4}$$

The factor of 2 from the $\pm k_\parallel$ symmetry cancels the $\frac{1}{2}$, giving

$$2\pi\, k_\perp\, \mathrm{d}k_\perp\, \mathrm{d}k_\parallel \;\longrightarrow\; \pi\, \frac{\mathrm{d}u\,\mathrm{d}v}{\sqrt{v}}, \qquad u \ge 0,\;\; v \ge 0. \tag{5}$$

In these variables the dispersions are *linear*:

$$\mathcal{E}_c(u,v) = \mathcal{E}_{0c} + A_c u - B_c v, \qquad \mathcal{E}_v(u,v) = -\mathcal{E}_{0v} - A_v u - B_v v. \tag{6}$$

### 2.3 Change of Variables III: Energy Space

We introduce two physically motivated variables:

$$\mathcal{E} \;\equiv\; \mathcal{E}_c(u,v), \qquad \Delta \;\equiv\; \mathcal{E}_c(u,v) - \mathcal{E}_v(u,v), \tag{7}$$

so that

$$\mathcal{E} - \mathcal{E}_{0c} = A_c\, u - B_c\, v, \tag{8}$$
$$\Delta - \mathcal{E}_g = \bar{A}\, u + \bar{B}\, v, \tag{9}$$

where

$$\boxed{\mathcal{E}_g \equiv \mathcal{E}_{0c} + \mathcal{E}_{0v}, \qquad \bar{A} \equiv A_c + A_v, \qquad \bar{B} \equiv B_v - B_c.} \tag{10}$$

#### Matrix Form and Jacobian

In matrix form:

$$\begin{pmatrix} \mathcal{E} - \mathcal{E}_{0c} \\ \Delta - \mathcal{E}_g \end{pmatrix} = \underbrace{\begin{pmatrix} A_c & -B_c \\ \bar{A} & \bar{B} \end{pmatrix}}_{M} \begin{pmatrix} u \\ v \end{pmatrix}. \tag{11}$$

The Jacobian determinant is

$$D \;\equiv\; \det M = A_c\bar{B} + B_c\bar{A} = A_c(B_v - B_c) + B_c(A_c + A_v) = \boxed{A_c B_v + A_v B_c \;>\; 0.} \tag{12}$$

Since all curvature coefficients are positive, $D > 0$ always. No sign ambiguity arises.

#### Inverse Transformation

$$\begin{pmatrix} u \\ v \end{pmatrix} = \frac{1}{D} \begin{pmatrix} \bar{B} & B_c \\ -\bar{A} & A_c \end{pmatrix} \begin{pmatrix} \mathcal{E} - \mathcal{E}_{0c} \\ \Delta - \mathcal{E}_g \end{pmatrix}, \tag{13}$$

giving explicitly

$$u(\mathcal{E},\Delta) = \frac{1}{D}\Big[\bar{B}\,(\mathcal{E} - \mathcal{E}_{0c}) + B_c\,(\Delta - \mathcal{E}_g)\Big], \tag{14}$$

$$v(\mathcal{E},\Delta) = \frac{1}{D}\Big[-\bar{A}\,(\mathcal{E} - \mathcal{E}_{0c}) + A_c\,(\Delta - \mathcal{E}_g)\Big]. \tag{15}$$

## 3 Integration Limits

The physical constraints $u \ge 0$ and $v \ge 0$ with $D > 0$ yield:

**From** $v \geq 0$ (eq. 15):

$$\bar{A}\,(\mathcal{E} - \mathcal{E}_{0c}) \;\leq\; A_c\,(\Delta - \mathcal{E}_g) \quad\Longrightarrow\quad \boxed{\mathcal{E} \;\leq\; \mathcal{E}_{0c} + \frac{A_c}{\bar{A}}\,(\Delta - \mathcal{E}_g) \;\equiv\; \mathcal{E}_{\max}(\Delta).} \tag{16}$$

**From** $u \geq 0$ (eq. 14):

$$\bar{B}\,(\mathcal{E} - \mathcal{E}_{0c}) \;\geq\; -B_c\,(\Delta - \mathcal{E}_g) \quad\Longrightarrow\quad \boxed{\mathcal{E} \;\geq\; \mathcal{E}_{0c} - \frac{B_c}{\bar{B}}\,(\Delta - \mathcal{E}_g) \;\equiv\; \mathcal{E}_{\min}(\Delta).} \tag{17}$$

*Physical interpretation:* $\mathcal{E}_{\max}$ corresponds to $v = 0$ ($k_\parallel = 0$, the transverse circle in $k$-space); $\mathcal{E}_{\min}$ corresponds to $u = 0$ ($k_\perp = 0$, along the symmetry axis). The integration domain $\mathcal{E}_{\min} \leq \mathcal{E} \leq \mathcal{E}_{\max}$ is non-empty for $\Delta > \mathcal{E}_g$, i.e., photon energies above the gap threshold.

## 4 Transformed Measure and Singularity Structure

From eq. (15), $v$ can be expressed in terms of $\mathcal{E}_{\max}$:

$$v(\mathcal{E}, \Delta) = \frac{\bar{A}}{D}\,\big(\mathcal{E}_{\max}(\Delta) - \mathcal{E}\big), \tag{18}$$

which is manifestly non-negative for $\mathcal{E} \leq \mathcal{E}_{\max}$. Substituting into the measure (5) and including the $1/D$ Jacobian:

$$\pi\,\frac{\mathrm{d}u\,\mathrm{d}v}{\sqrt{v}} \;\longrightarrow\; \frac{\pi}{\sqrt{\bar{A}\,D}}\;\frac{\mathrm{d}\Delta\;\mathrm{d}\mathcal{E}}{\sqrt{\mathcal{E}_{\max}(\Delta) - \mathcal{E}}}. \tag{19}$$

The integrable singularity $(\mathcal{E}_{\max} - \mathcal{E})^{-1/2}$ occurs at the **upper limit**, where $v = 0$ (the transverse circle). This upper-limit singularity is the hallmark of the saddle-point ($M_1$) topology and applies identically at both X and L.

## 5 Resolution of the $\delta$-Function

Substituting eq. (19) into eq. (1) and using $\mathcal{E}_v = \mathcal{E} - \Delta$:

$$\varepsilon_2(\hbar\omega) \propto \frac{|P|^2}{(\hbar\omega)^2\sqrt{\bar{A}\,D}} \iint f(\mathcal{E} - \Delta)\,\big[1 - f(\mathcal{E})\big]\;\delta(\Delta - \hbar\omega)\;\frac{\mathrm{d}\Delta\;\mathrm{d}\mathcal{E}}{\sqrt{\mathcal{E}_{\max}(\Delta) - \mathcal{E}}}. \tag{20}$$

The $\delta$-function sets $\Delta = \hbar\omega$ and eliminates one integration, yielding the central result:

$$\boxed{\varepsilon_2(\hbar\omega) = \frac{\mathcal{C}\,|P|^2}{(\hbar\omega)^2\,\sqrt{\bar{A}\,D}} \int_{\mathcal{E}_{\min}(\hbar\omega)}^{\mathcal{E}_{\max}(\hbar\omega)} \frac{f(\mathcal{E} - \hbar\omega)\;\big[1 - f(\mathcal{E})\big]}{\sqrt{\mathcal{E}_{\max}(\hbar\omega) - \mathcal{E}}}\;\mathrm{d}\mathcal{E},} \tag{21}$$

with

$$\mathcal{E}_{\max}(\hbar\omega) = \mathcal{E}_{0c} + \frac{A_c}{\bar{A}}\,(\hbar\omega - \mathcal{E}_g), \qquad \mathcal{E}_{\min}(\hbar\omega) = \mathcal{E}_{0c} - \frac{B_c}{\bar{B}}\,(\hbar\omega - \mathcal{E}_g), \tag{22}$$

where $\mathcal{C}$ absorbs the universal prefactor $8\pi^3 e^2 \hbar^4/(3m^2)$.

*For emission* (conduction $\to$ valence), the Fermi factor becomes $f(\mathcal{E})\,[1 - f(\mathcal{E} - \hbar\omega)]$; at equilibrium the two are related by detailed balance: $\varepsilon_2^{\mathrm{em}}/\varepsilon_2^{\mathrm{abs}} = e^{-\beta\hbar\omega}$.

## 6 Physical Onset: X vs. L

Although the integral structure (21) is identical at both critical points, their physical behaviour near onset differs because of the position of $\mathcal{E}_{0c}$ relative to the Fermi level.

### X-Point: Empty Conduction Band

At the X point, $\mathcal{E}_{0c}^X = 0$ (at $E_F$, by convention). The conduction band minimum sits at or above the Fermi level: the final states are *empty*.

**Geometric sub-gap tail.** The saddle topology permits momentum-conserving transitions even for $\hbar\omega < \mathcal{E}_g$, because the $-B_c v$ term allows $\Delta(\mathbf{k}) < \mathcal{E}_g$ at finite $v$. This generates a smooth sub-gap tail that decays exponentially as the final states are pushed deep below $E_F$, where $1 - f(\mathcal{E}) \to 0$.

**Practical lower bound.** The geometric $\mathcal{E}_{\min}$ lies far below $E_F$ where the Fermi factor exponentially suppresses the integrand. It is therefore replaced by a thermal cutoff:

$$\mathcal{E}_{\min}^X \approx -20\, k_B T,$$

which safely truncates the integral without affecting physical accuracy.

### L-Point: Submerged Minimum

At the L point, $\mathcal{E}_{0c}^L < 0$ — the conduction band minimum is *submerged below* the Fermi level due to the Fermi surface "neck" at the BZ boundary.

**Pauli-blocked onset.** Although transitions are geometrically allowed for all $\hbar\omega > \mathcal{E}_g$, those near the geometric threshold terminate in states below $E_F$ that are already occupied. The physical onset occurs at the elevated energy $\hbar\omega_{\text{onset}} \approx E_F - \mathcal{E}_{0v}^L$ where final states emerge above $E_F$.

**Practical lower bound.** The geometric $\mathcal{E}_{\min}^L$ again lies deep below $E_F$ (even more so, since $\mathcal{E}_{0c}^L < 0$). The same thermal cutoff applies:

$$\mathcal{E}_{\min}^L \approx -20\, k_B T,$$

and the onset shape is governed by the smooth sigmoidal form of $[1 - f(\mathcal{E})]$ rather than by any geometric bound.

In both cases, the onset is physically controlled by the Fermi–Dirac factor $[1 - f(\mathcal{E})]$ rather than by the geometric integration limits. The sub-gap tail (X) and the Pauli-delayed onset (L) are both natural consequences of the same thermal weighting — no piecewise bounds or *ad hoc* broadening are needed.

## 7 Numerical Regularization

The singularity at $\mathcal{E} = \mathcal{E}_{\max}$ is integrable but numerically inconvenient. The substitution

$$t = \sqrt{\mathcal{E}_{\max} - \mathcal{E}}, \qquad \mathrm{d}\mathcal{E} = -2t\,\mathrm{d}t, \qquad \mathcal{E} = \mathcal{E}_{\max} - t^2, \tag{23}$$

transforms (21) into a smooth integral over a uniform $t$-grid:

$$\boxed{\varepsilon_2(\hbar\omega) = \frac{2\,\mathcal{C}\,|P|^2}{(\hbar\omega)^2\sqrt{\bar{A}\,D}} \int_0^{\sqrt{\mathcal{E}_{\max}-\mathcal{E}_{\min}}} f\big(\mathcal{E}_{\max} - t^2 - \hbar\omega\big)\,\big[1 - f\big(\mathcal{E}_{\max} - t^2\big)\big]\,\mathrm{d}t,} \tag{24}$$

evaluated by Simpson quadrature. This form is used for both X and L.

## 8 Verification: Zero-Temperature JDOS

At $T = 0$ the Fermi functions become step functions. For $\hbar\omega$ just above threshold:

$$J(\hbar\omega) \propto \int_{\mathcal{E}_{\min}}^{\mathcal{E}_{\max}} \frac{\mathrm{d}\mathcal{E}}{\sqrt{\mathcal{E}_{\max} - \mathcal{E}}} = 2\sqrt{\mathcal{E}_{\max} - \mathcal{E}_{\min}} \;\propto\; \sqrt{\hbar\omega - \mathcal{E}_g}, \tag{25}$$

recovering the $M_0$-type onset ($\sqrt{\cdot}$ threshold). While each individual band is a saddle ($M_1$), the *energy-difference* surface $\Delta(u, v) = \mathcal{E}_g + \bar{A}u + \bar{B}v$ is a minimum in $(u, v)$ space (both $\bar{A}, \bar{B} > 0$ for $B_v > B_c$), giving $M_0$ onset for the JDOS.

## 9 Full Model for Gold

Since both critical points share the same integral structure (24), the full interband $\varepsilon_2$ is a weighted sum:

$$\varepsilon_2(\omega) = \frac{S}{\omega^2}\Big[\big|P_X/P_L\big|^2 N_X\, J_X(\omega, T) \;+\; N_L\, J_L(\omega, T)\Big] + \frac{A_D}{\omega^3}, \tag{26}$$

where $J_{X,L}$ are evaluated using (24) with the respective parameters:

Table 2: Derived quantities. The formulas are identical at both points; only the numerical values differ.

| Quantity | **X point** | **L point** |
|---|---|---|
| Multiplicity $N$ | 6 | 8 |
| $\mathcal{E}_{0c}$ | 0 (reference) | fitted ($< 0$, submerged) |
| $\mathcal{E}_g$ | $\sim$1.94 eV | $\sim$2.45 eV |
| $\bar{A} = A_c + A_v$ | $A_c^X + A_v^X$ | $A_c^L + A_v^L$ |
| $\bar{B} = B_v - B_c$ | $B_v^X - B_c^X$ | $B_v^L - B_c^L$ |
| $D = A_cB_v + A_vB_c$ | $> 0$ | $> 0$ |
| Singularity | $(\mathcal{E}_{\max} - \mathcal{E})^{-1/2}$ (upper limit) | |
| $\mathcal{E}_{\max}(\hbar\omega)$ | $\mathcal{E}_{0c} + \dfrac{A_c}{A_c + A_v}(\hbar\omega - \mathcal{E}_g)$ | |
| $\mathcal{E}_{\min}(\hbar\omega)$ | $\approx -20\, k_BT$ (thermal cutoff) | |

The **only** differences between X and L are:

1. Different effective masses $(m^*_{c\perp}, m^*_{c\|}, m^*_{v\perp}, m^*_{v\|})$.
2. Different gap energies $\mathcal{E}_g$.
3. $\mathcal{E}^L_{0c} \neq 0$ (submerged below $E_F$), whereas $\mathcal{E}^X_{0c} = 0$ by convention.
4. Different BZ multiplicities ($N_X = 6$, $N_L = 8$).

## Summary of Transformations

$$\underbrace{d^3k}_{\text{3D BZ}} \xrightarrow{\text{cyl.}} \underbrace{2\pi k_\perp\, dk_\perp\, dk_\|}_{\text{2D}} \xrightarrow{u,v} \pi\underbrace{\frac{du\,dv}{\sqrt{v}}}_{\text{linear}} \xrightarrow{\mathcal{E},\Delta} \underbrace{\frac{\pi}{\sqrt{\bar{A}D}}\frac{d\Delta\, d\mathcal{E}}{\sqrt{\mathcal{E}_{\max} - \mathcal{E}}}}_{\text{energy space}} \xrightarrow{\delta} \underbrace{\text{1D}}_{(21)}$$

## 10 Python Implementation

The code below implements eq. (24) for both critical points using the $t$-substitution and Simpson quadrature. The Fermi weight uses the numerically stable log-sum-exp form.

```python
import numpy as np
from scipy.integrate import simpson

kB    = 8.617e-5    # Boltzmann constant [eV/K]
C     = 3.81        # hbar^2 / (2 m_e)  [eV*Ang^2]
N_X   = 6           # X-point multiplicity
N_L   = 8           # L-point multiplicity
N_INT = 80          # Simpson quadrature points

def fermi_weight(E, hw, T):
    """Absorption weight f(E-hw)*[1-f(E)], numerically stable."""
    if T == 0:
        return ((E - hw) < 0).astype(float) * (E >= 0).astype(float)
    beta = 1.0 / (kB * T)
    return np.exp(
        -np.logaddexp(0, beta * (E - hw))
        -np.logaddexp(0, -beta * E)
    )

def compute_model(p, hw_grid):
    """Return (e2_X, e2_L, e2_XL, e2_XLD) from parameter dict p."""
    # Curvature coefficients from effective masses
    Ac_X, Bc_X = C / p['mc_perp_X'], C / p['mc_par_X']
    Av_X, Bv_X = C / p['mv_perp_X'], C / p['mv_par_X']
    Ac_L, Bc_L = C / p['mc_perp_L'], C / p['mc_par_L']
    Av_L, Bv_L = C / p['mv_perp_L'], C / p['mv_par_L']

    # Both X and L: Ec = E0c + Ac*k_perp^2 - Bc*k_par^2  (saddle)
    Abar_X = Ac_X + Av_X
    D_X    = Ac_X * Bv_X + Av_X * Bc_X   # always > 0
    Abar_L = Ac_L + Av_L
    D_L    = Ac_L * Bv_L + Av_L * Bc_L   # always > 0

    pf_X = 1.0 / np.sqrt(Abar_X * D_X)
    pf_L = 1.0 / np.sqrt(Abar_L * D_L)

    T     = p['T']
    Eg_X  = p['Eg_X']
    Eg_L  = p['Eg_L']
    E0c_L = p['E0c_L']    # submerged below E_F (negative)

    def jdos(w, Abar, Ac, pf, Eg, E0c):
        """Regularized 1D integral via t = sqrt(E_max - E)."""
        e_max = E0c + (Ac / Abar) * (w - Eg)
        e_min = -20.0 * kB * T if T > 0 else -0.1
        if e_max <= e_min:
            return 0.0
        t = np.linspace(0, np.sqrt(e_max - e_min), N_INT)
        # E = e_max - t^2;  dE = 2t dt  =>  singularity cancelled
        return pf * simpson(
            2.0 * fermi_weight(e_max - t**2, w, T), x=t
        )

    S = p['Scale']
    e2_X = S * np.array([
        N_X * p['P_ratio_sq']
        * jdos(w, Abar_X, Ac_X, pf_X, Eg_X, 0.0) / w**2
        for w in hw_grid
```

```
    ])
    e2_L = S * np.array([
        N_L * jdos(w, Abar_L, Ac_L, pf_L, Eg_L, E0c_L) / w**2
        for w in hw_grid
    ])
    return e2_X, e2_L, e2_X + e2_L, \
           e2_X + e2_L + p['A_drude'] / hw_grid**3
```

Listing 1: Rosei interband $\varepsilon_2$ — regularized Simpson integration

**References:** M. Guerrisi, R. Rosei, and P. Winsemius, *Phys. Rev. B* **12**, 557 (1975). N. E. Christensen and B. O. Seraphin, *Phys. Rev. B* **4**, 3321 (1971).